

1. Four key steps in a data science study

1. Data preparation
2. Algorithm selection
3. Parameter tuning
4. Model evaluation

2. k-means clustering

k-means clustering is an unsupervised learning method that groups similar data points into k *clusters* based on distance from cluster centers.

Works best when:

- Clusters are **clear, separate, and spherical**
- Data is **numeric and structured**
- Number of clusters (k) is known or can be estimated

3. PCA (Principal Component Analysis)

PCA is a **dimension reduction technique** that transforms many variables into fewer new variables (principal components) that capture the most variation in the data.

Works best when:

- Data has **many correlated variables**
- Goal is **simplification or visualization**
- You want to reduce noise while keeping important patterns

4. Three common ways to measure associations

1. **Support** – how often items occur together
2. **Confidence** – probability of one item given another
3. **Lift** – strength of association compared to random chance

5. Louvain vs PageRank

Louvain Method

Detects **communities/clusters** in networks

PageRank

Measures **importance of nodes**

Louvain Method

Groups similar nodes together

Used for segmentation

Focus = **structure**

PageRank

Ranks nodes based on connections

Used for influence ranking

Focus = **importance**

6. Drawbacks of regression analysis

- Assumes **linear relationships**
- Sensitive to **outliers**
- Can suffer from **overfitting**
- Requires **good data quality**
- Cannot capture **complex non-linear patterns** easily

7. Unsupervised learning algorithms

k-means clustering

- Principal Component Analysis (PCA)
- Association rules
- Social network analysis

8. Supervised learning algorithms + tuning parameters

Algorithms:

- Regression
- k-Nearest Neighbors (k-NN)
- Support Vector Machine (SVM)
- Decision Tree
- Random Forest
- Neural Networks

Common tuning parameters:

- k (for k-NN, k-means)
- Tree depth (Decision Tree)
- Number of trees (Random Forest)
- Learning rate (Neural Networks)
- Regularization parameter (Regression/SVM)